

Sigenenergy OpenAPI

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Sigenergy Openapi, launched by Sigenergy, is a software product designed to facilitate device-related interactions for vendors. This document primarily outlines the functionalities provided by Sigenergy Openapi for vendors and the specifications of interaction processes between Sigenergy Openapi and vendors. In order to meet the requirements of vendors to the fullest extent, Sigenergy Openapi offers the following functionalities:

The vendor can register the power system that need to be managed to the service through the Sigenergy Openapi. After registration, the service can interact with the system through the Sigenergy Openapi. Registration includes necessary authentication permissions and system integration. Please refer to [Authentication](#) and [Boarding](#) for details.

The Sigenergy Openapi can periodically push telemetry data of system to the service. To meet the requirements of vendors, Sigenergy Openapi offers customizable push intervals and customizable push content within optional ranges. Please contact Sigenergy to customize the telemetry push frequency and data content required. Please refer to [Telemetry](#) for details.

For certain information about system, such as battery capacity and operational status, which do not frequently change, the Sigenergy Openapi will provide them during system registration. Additionally, whenever there is a change in this information, the Sigenergy Openapi will push the updated information to vendors to ensure that the vendors can promptly understand the system status. Similar to Telemetry data, please contact Sigenergy to communicate the required system data content for push. Please refer to [System Data](#) for details.

Some information about the system is crucial, such as system failure and recovery. vendors may be highly interested in this information, so the Sigenergy Openapi provides an alarm push feature. Similarly, the content of alarms is customizable, and please contact Sigenergy. Please refer to [Alarm](#) for details.

Please note that service providers need to provide three separate links for receiving telemetry, system, and alarm data. Currently, these three links cannot be configured automatically and require support from Sigenergy.

The Sigenergy Openapi provides services with the capability to issue commands to system. Through command issuance, services can manage charging and

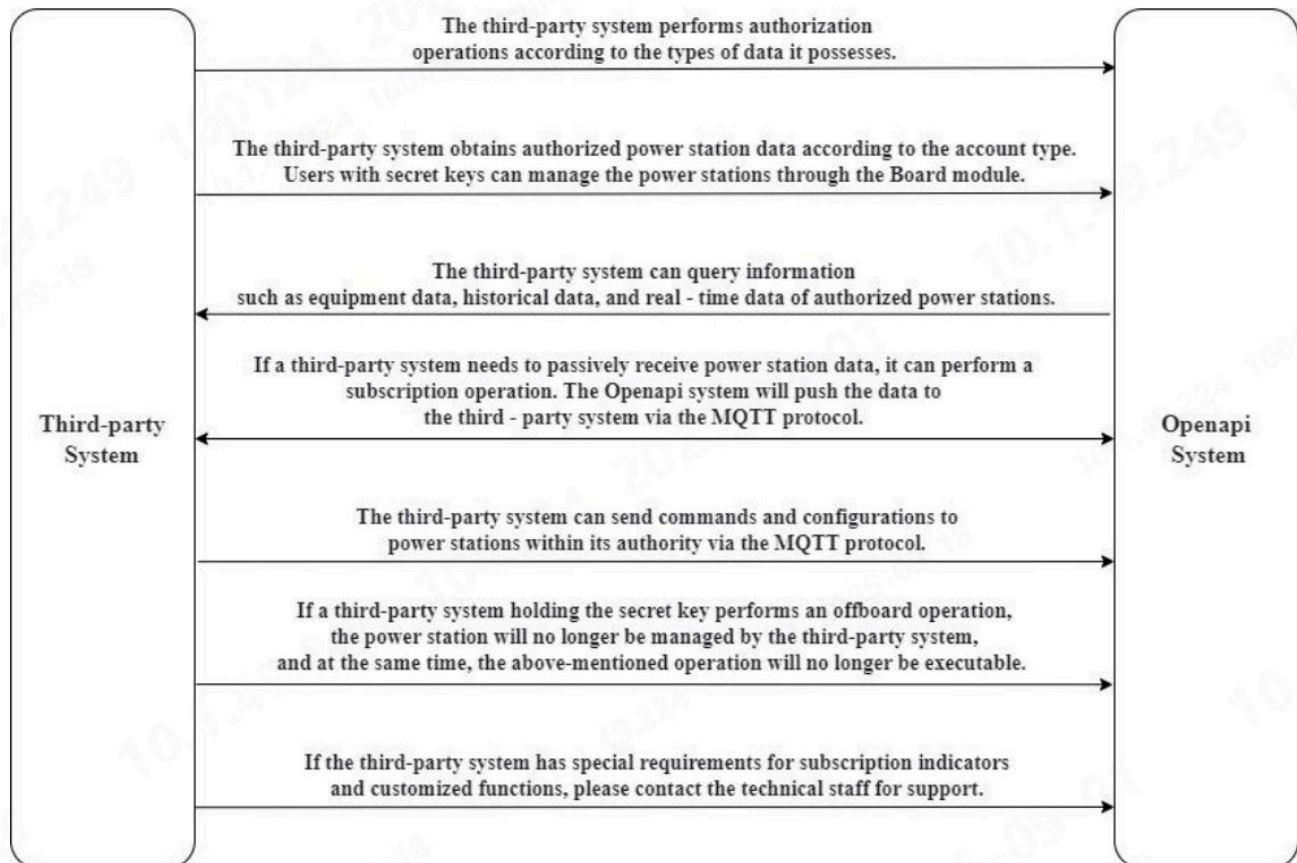
discharging strategies of system according to their own requirements. Command issuance is a highly flexible feature, and Sigenenergy currently offers some commonly used commands, such as controlling system for forced charging or discharging. Please refer to [Instruction](#) for details.

Definitions And Abbreviations

Term	Definition
VPP	Virtual Power Plant.
NBI	Northbound
API	Application Programming Interface.
systemId	The unique code representing the power station.
onboard	System is registered with the cloud management, allowing vendors to receive system information and control system by issuing commands.
offboard	System is disassociated from cloud management, vendors will be unable to control the system.
PV	Photovoltaic.
snCode	Serial number code.
system	A storage system composed of inverters, batteries, gateways, and other physical components.
MQTT	Message Queuing Telemetry Transport

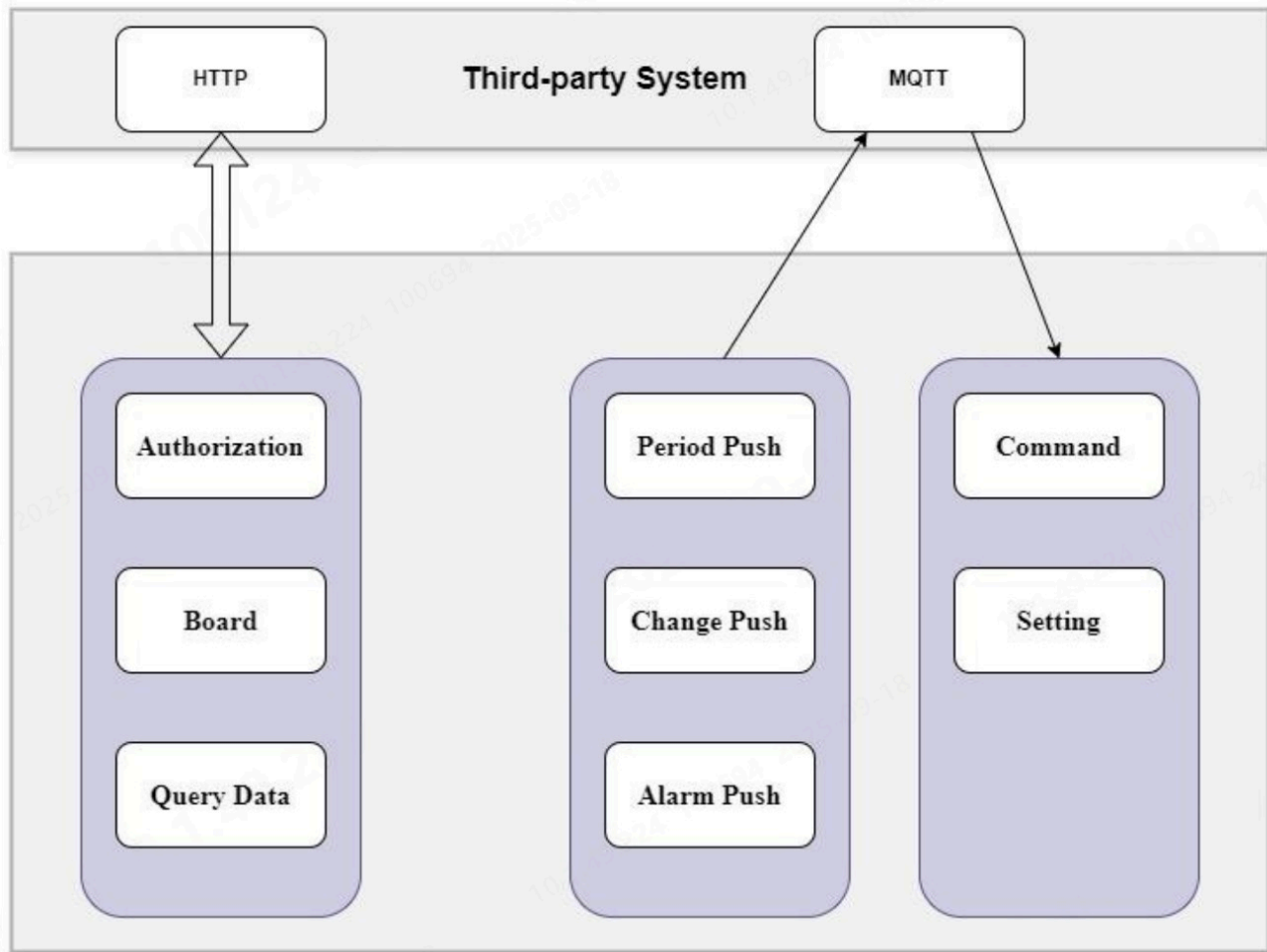
System Architecture

Currently, the interaction between third-party systems and the **OpenAPI Service** is mainly implemented through **HTTP protocol** and **MQTT protocol**:



HTTP Protocol: Suitable for **synchronous request scenarios**, such as **authentication**, **user registration**, and **data query** modules. The caller can obtain the execution result immediately after the request is completed.

MQTT Protocol: Suitable for **asynchronous message scenarios**, such as **data push** and **command control** modules. The OpenAPI service can also proactively push data to third-party systems via the MQTT protocol.



Rate Limiting Policy

To ensure the rational use of system resources, improve overall service efficiency, and effectively defend against external malicious traffic attacks, the OpenAPI Service defines rate limiting rules based on the expected access scale.

For requests beyond the expected range, the system may take measures such as **service rejection**, **request queuing**, or **service degradation**.

The current rate limiting policies are as follows:

Device Quantity Limit per Request

The number of devices involved in a single API request must be fewer than **200 units**.

API Access Frequency Limit

Depending on the business characteristics and resource consumption, the platform sets differentiated frequency limits for different APIs. Detailed rules are specified in the corresponding API documentation.

If you need to adjust the access frequency limit, please contact the technical support team.

General Notes

Domain

All API requests must use the specified domain: `openapi-eu.sigencloud.com`.

Topic

The OpenAPI Service allocates dedicated **Topics** for third-party systems based on business types and access configurations.

The specific Topic can be viewed in the **Data Subscription** section of the application in the Open Platform Console.

Topics for receiving commands and configuration have been clearly defined in the corresponding documents. Please refer to:

[Subscription](#)

[Instruction](#)

Request

All API interfaces are based on the **HTTP protocol**.

The request header must include the **Authorization** parameter for permission validation.

Request parameters must follow the examples provided in the API documentation.

Response Body

All responses are in **standard JSON format** and can be parsed using common JSON libraries.

For HTTP responses, the response header **Content-Type** must be set to `application/json`.

Response Body Field Description:

Name	Type	Description
code	Integer	Status code
msg	String	Description of the status
timestamp	Long	Response generation time (s)
data	Object	Data content returned by API

Introduction

The current system supports two scenarios, namely the Virtual Power Plant scenario and the Northbound scenario. The specific introductions are as follows:

VPP (Virtual Power Plant) Scene: In this scene, the equipment is controlled by the service provider, and the owner has no right to switch the operation mode of the power plant. At the same time, the system supports the service provider to subscribe to and actively query the data of the power plant.

NBI (North Bound) Scene: In this scene, both the owner and the service provider can actively switch the operation mode of the power plant. When the service provider needs to control the power plant, it can switch to the Northbound mode. If the owner wishes to switch to the Time-of-Use (TOU) mode or the full-scale power grid connection mode, the operation can also be carried out. In addition, the system also supports the service provider to actively query the data.

Supplementary Notes on VPP Scene

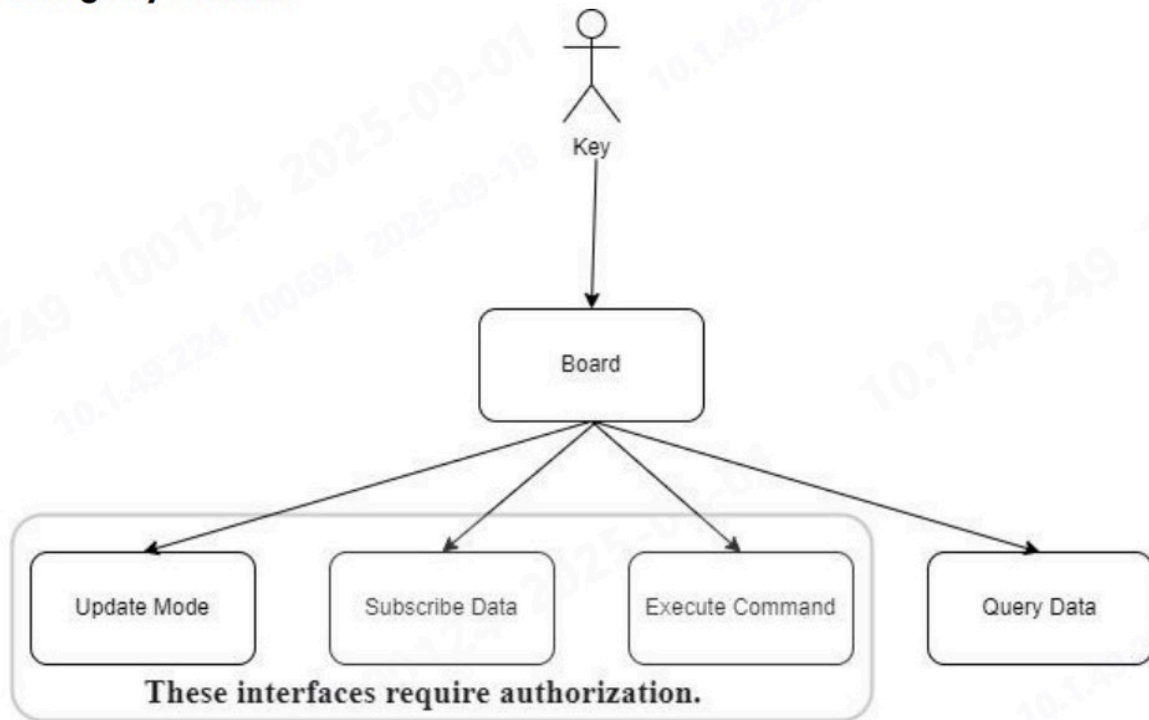
At present, the VPP mode is only supported for use by service providers who possess the secret key. Once the service provider executes the onboard interface, they will acquire the operational authority over the power station, and this operation will also switch the power station to run in the VPP mode. In this mode, the owner has no right to switch to other modes. After the onboard operation is carried out, if the service provider requires the device to report the operation data, they can subscribe to the device data (for specific operation steps, please refer to [Subscription](#)). Upon receiving the subscription information from the service provider, the power station will report the corresponding data according to the subscribed data type. Detailed descriptions of different subscription types can be found in [Telemetry](#), [System Data](#), and [Alarm](#). Moreover, the service provider can also issue commands targeting the power station, and the specific content is elaborated in [Instruction](#). If the service provider needs to actively query the data of the power station, they can refer to the relevant interface instructions in [Inventory](#), [Historical Data](#), and [RealTime Data](#). The owner cannot actively exit the VPP scenario, and it is necessary for the service provider to perform the exit operation. Once the VPP scenario is exited, the power station will depart from the VPP

mode and switch to the maximum self-generation and self-consumption mode. After exiting the VPP scenario, the service provider will no longer be able to carry out any of the operations mentioned above on the power station.

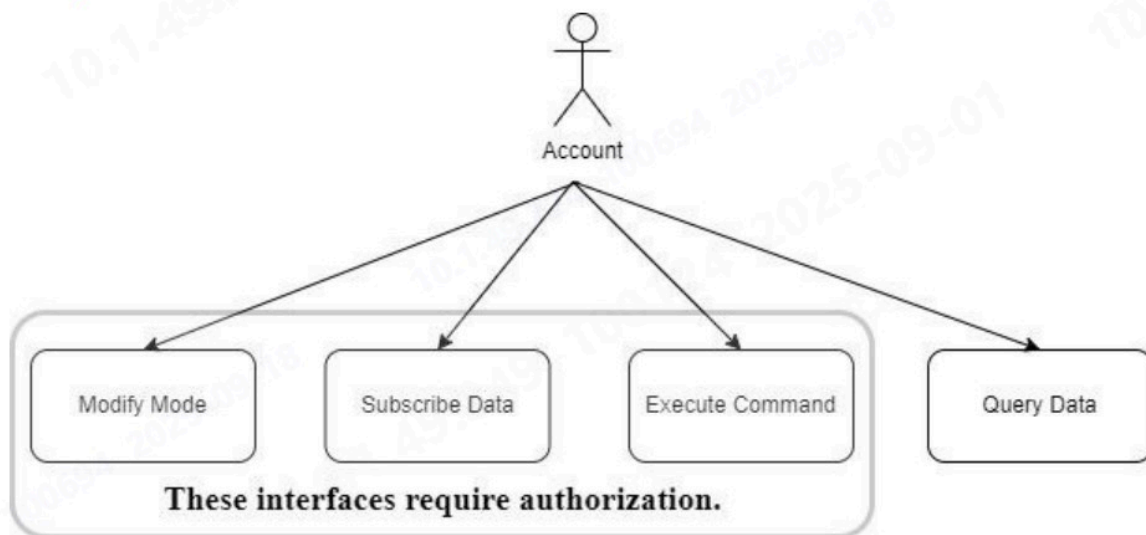
Supplementary Notes on NBI Scene

The Northbound scenario supports two methods for obtaining tokens: using a secret key and using an account and password. The main difference between the two methods lies in the power station authorization process. Customers using a secret key must go through the onboard authorization process, while those using an account and password do not need to execute the onboard process. For specific differences, please refer to the following figure. It should be noted that in the Northbound scenario, only the query of power station data is supported by default. For other functions, you need to apply to technical personnel for access permissions.

1. Using Key Provider



2. Using Account Provider



Introduction

Before accessing the Openapi service, users need to apply to technical staff for an authorization code. Each time they call the API, they must use this authorization code to request a token.

When a client application requests a (client) token from the authorization server, the request must include specific parameters and be encoded in the "application/x-www-form-urlencoded" format.

If a third-party user program wants to retrieve data from Sigenergy Cloud, it must implement the authentication and authorization process, which includes the following details:

Sigenergy Cloud will grant third-party users a client ID (`client_id`) and a client secret (`client_secret`) so that they can obtain tokens in combination with the standard OAuth2 client authorization types.

The third-party user program needs to regularly obtain the authorization decision from the resource owner. Only after the authorization is approved can it access user resources.

The default expiration time for a token is 12 hours. Once the token expires, you need to obtain a new token again through the client authorization type.

Access Restriction

Password Error Locking Rule: Within a 60-minute period, if a user enters an incorrect password consecutively for five times, the user will be locked out for 3 minutes.

When a user is locked due to this rule, the API will return the error code 11002;

Access Rate Limit for Third-Party Platform Users: Each third-party platform user is restricted to a maximum of 10 accesses per minute;

Authorization Rule for Third-Party User Programs: Third-party user programs need to periodically obtain the authorization decision of the resource owner. Only after the authorization is approved can they access user resources.

Request Parameter

Name	Type	Required	Description
username	String	Yes	Username used during user registration
password	String	Yes	Password

Response Message

Name	Type	Description
code	Integer	Error Code
msg	String	Message
data	String	The data contains the following contents
>accessToken	String	Access token
>expiresIn	Long	Expiration time of the token

Name	Type	Description
>tokenType	String	Token type

Sample Request

```
{  
  "username": "test@test.com",  
  "password": "kNlsj6voh+Q"  
}
```

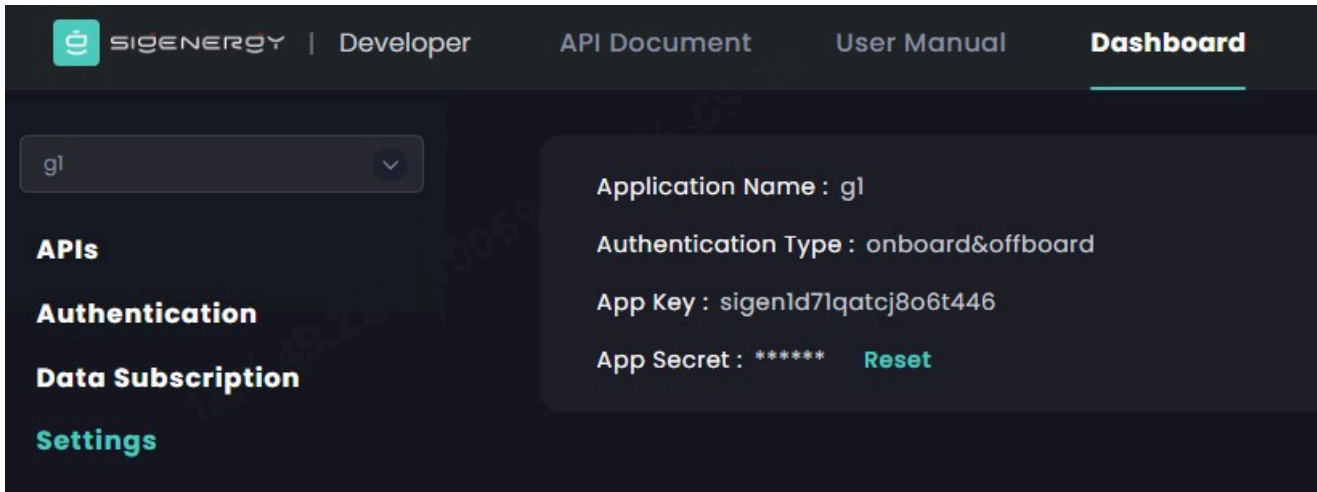
Sample Successful Response

```
{  
  "code": 0,  
  "msg": "success",  
  "data": "  
{"tokenType": "Bearer", "accessToken": "HgrU1Rn2CVUx4rV8C7zpEIFBYz1gUW8kg6VDv  
9kR5oMtTrzXeKVopndPJWdd9foRSxrqpRGXC5ykHz05W5pXIhWVsAExbPM3i7p2UpLsXTl3ovf  
oFEwHHZRLJDMFmB", "expiresIn": 43199}"  
"
```

Sample Failed Response

```
{  
  "code": 11003,  
  "msg": "authentication failed",  
  "data": null  
}
```

Authentication



In the **Control Center** → **Settings** page, you can obtain the **Application Key (App Key)** and **Application Secret (App Secret)**. The platform follows the **OAuth2 standard protocol, Client Credentials Grant type** for authentication.

You need to concatenate AppKey:AppSecret, encode it with Base64:
`base64(AppKey:AppSecret)`; Then the result would be placed in the request body field key. Then send a request to the authentication server's Token endpoint to obtain the access token (accessToken) required for calling subsequent APIs.

Access Limitations

For third-party platform users, the access frequency is limited to **a maximum of 10 requests per minute**.

Request Parameters

Name	Type	Required	Description
key	String	Yes	Obtain it by requesting from Sigenenergy.

Response Message

Name	Type	Description
code	Integer	Error code
msg	String	Message
data	String	Data, includes:
>accessToken	String	Access token
>expiresIn	Long	Token expiration
>tokenType	String	Token type

Request Body Example

```
{
  "key": "aslsjsvoh9Qq0"
}
```

Successful Response Example

```
{
  "code": 0,
  "msg": "success",
  "data": "
```



```
{\"tokenType\": \"Bearer\", \"accessToken\": \"HgrU1Rn2CVUx4rV8C7zpEIFBYz1gUW8kg6VDv9kR5oMtTrzXeKVopndPJWdd9foRSxrqpRGXC5ykhz05W5pXIhWVsAExbPM3i7p2UpIsXTl3ovfoFEwHHZRLJDMFmB\", \"expiresIn\": 43199}"}
```

Failed Response Example

```
{  
  "code": 11003,  
  "msg": "authentication failed",  
  "data": null  
}
```

The Board module is a module for customers with keys to manage the power station. The onboard interface is a process for authorizing the power station to the service provider, which is also a prerequisite for subsequent processes.

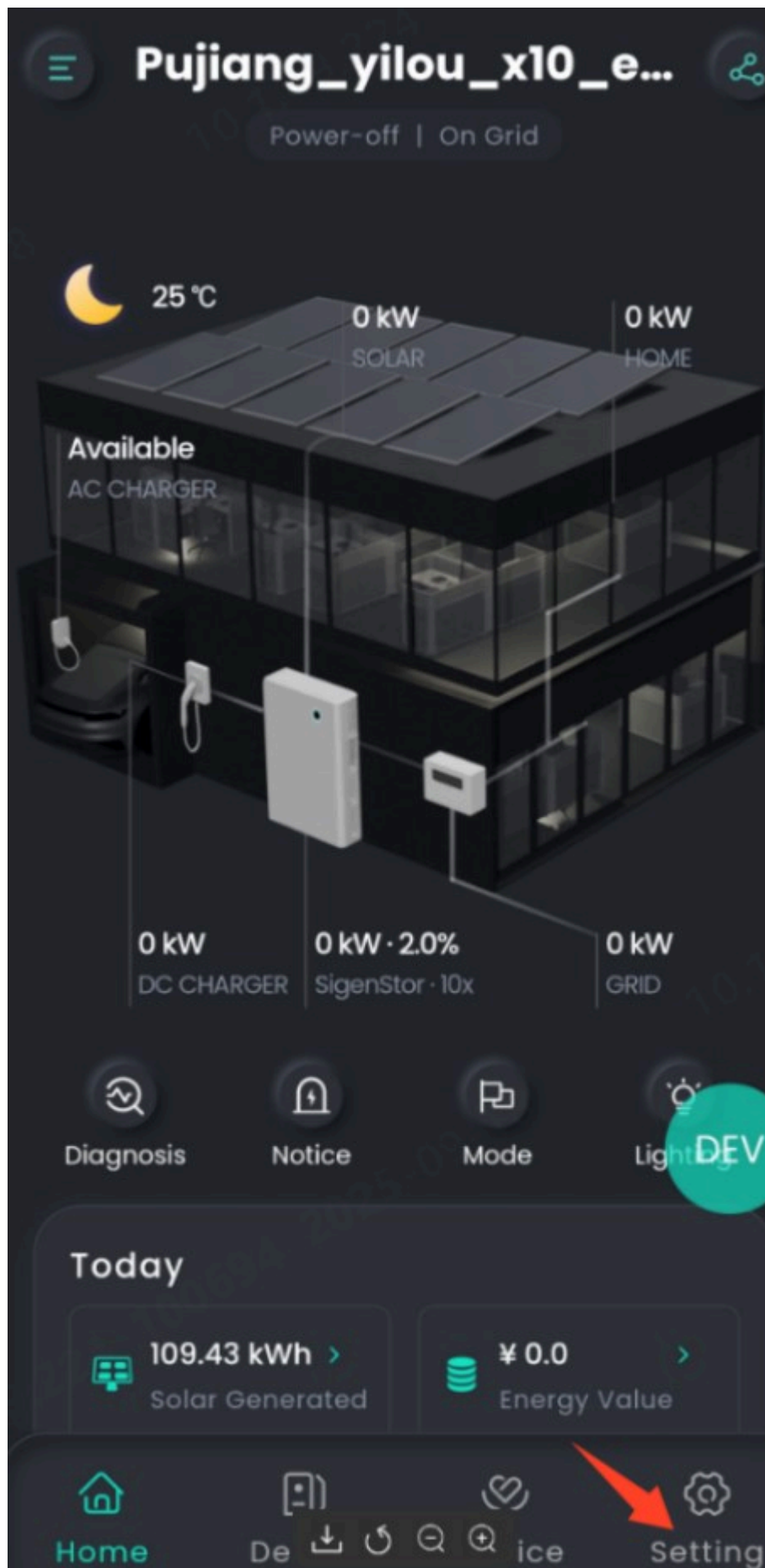
The offboard interface is a process for revoking the authorization of the power station from the service provider. After the permission is revoked, the service provider will be unable to operate the power station and will not be able to receive the data of the power station.

It is important to note that the `systemId` used in the API interfaces in this chapter refers to the unique code representing the power station.

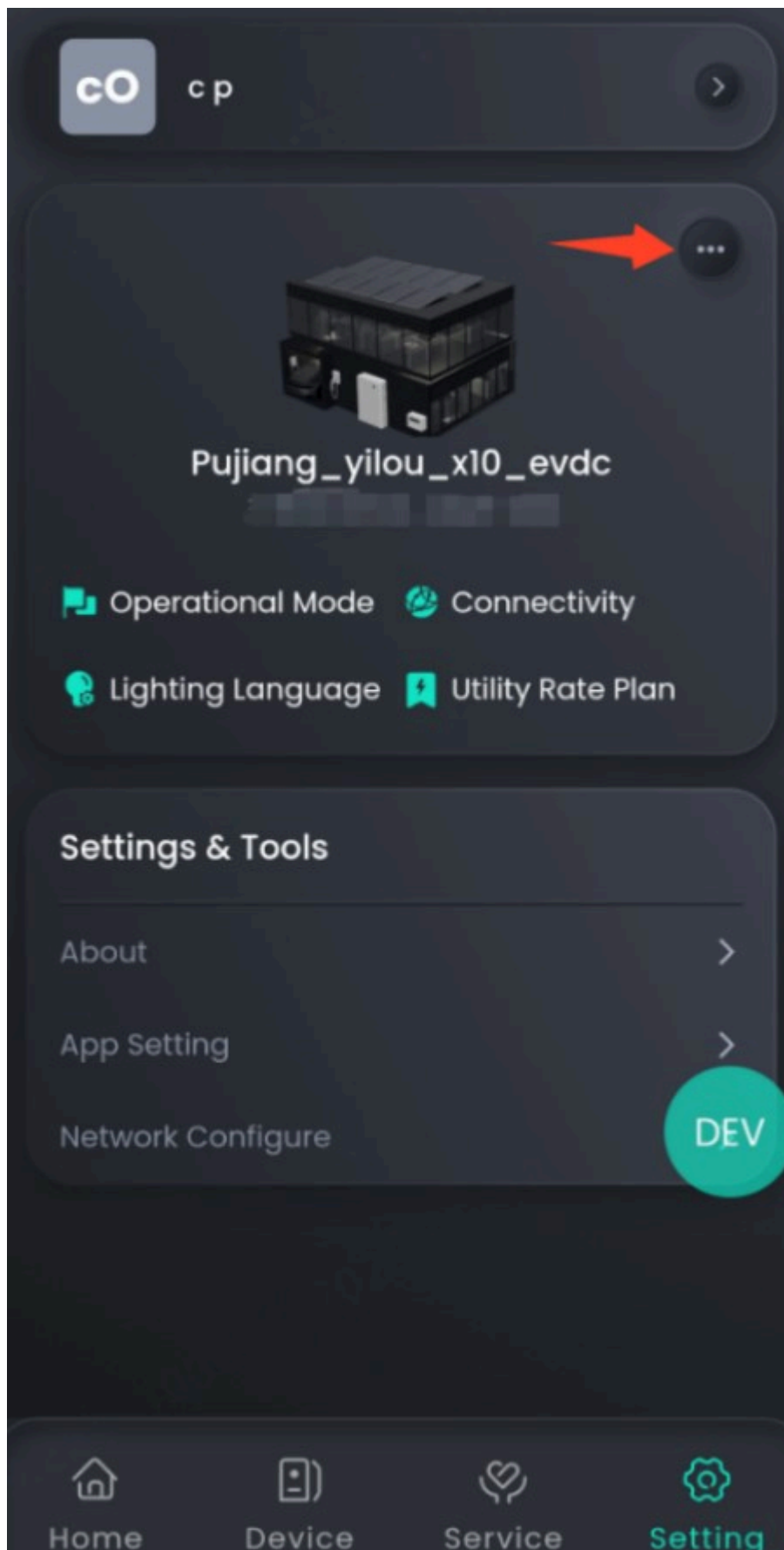
Boarding > Obtain the SystemId

Sigenenergy Overview 2026-07-16 10:00

Step 1: In the owner interface, click the Setting button.



Step 2: Click the icon in the upper right corner, which looks like three dots.



Step 3: The systemId information for the power station will be displayed in the Basic Info section.

System Settings



Pujiang_yilou_x10_evdc

Basic Info

System ID

KXGCSI727160960

Country/Region

CN

Time Zone

(UTC+08:00)Shanghai

Grid Connection Date

2024-09-24

Total Panel Capacity

250.0 kWp

Battery Capacity

314.56 kWh

This API is used to onboard a specified system (systemId) into the platform.

Request Headers

Header	Required	Type	Description
Authorization	Yes	String	Authentication token, format: Bearer <access_token>
Content-Type	Yes	String	application/json

```
Authorization: Bearer D9gwcA4L6D-j2tM9jnYsYTeDM
Content-Type: application/json
```

Request Body

Type: application/json Supports sending multiple system IDs in a batch.

Example:

```
[
  "KXGCS1727160960",
  "PGIYT1142977051"
]
```

Response Body

```
{
  "code": 0,
  "msg": "success",
}
```

```
{
  "timestamp": 1757580062,
  "data": [
    {
      "systemId": "KXGCS1727160960",
      "result": true,
      "codeList": [
        0
      ]
    },
    {
      "systemId": "PGIYT1142977051",
      "result": false,
      "codeList": [
        1502
      ]
    }
  ]
}
```

Response Parameters

Field	Type	Description
code	int	Response status code (0 indicates success, non-zero indicates failure)
msg	string	Response message
timestamp	long	UNIX timestamp (seconds)
data	array	Array of onboarding results

This API is used to offboard (remove) a specified system (systemId) from the platform.

Request Headers

Header	Required	Type	Description
Authorization	Yes	String	Authentication token, format: Bearer <access_token>
Content-Type	Yes	String	application/json

```
Authorization: Bearer D9gwcA4L6D-j2tM9jnYsYTeDM
Content-Type: application/json
```

Request Body

Type: application/json Supports sending multiple system IDs in a batch.

Example:

```
[
  "KXGCS1727160960",
  "PGIYT1142977051"
]
```

Response Body

```
{
  "code": 0,
  "msg": "success",
}
```

```
{
  "timestamp": 1757580182,
  "data": [
    {
      "systemId": "KXGCS1727160960",
      "result": true,
      "codeList": [
        0
      ]
    },
    {
      "systemId": "PGIYT1142977051",
      "result": true,
      "codeList": [
        0
      ]
    }
  ]
}
```

Response Parameters

Field	Type	Description
code	int	Response status code (0 indicates success, non-zero indicates failure)
msg	string	Response message
timestamp	long	UNIX timestamp (seconds)
data	array	Array of onboarding results

Onboard / Offboard Result Codes

/board/onboard and /board/offboard support batch processing for multiple `systemId` values. The top-level code indicates whether the API request itself was processed by the platform. For each site, check `data[].result` and `data[].codeList`.

Different sites in the same batch request may have different results. Do not use only the top-level `code=0` to assume that all sites were onboarded or offboarded successfully.

How to Read a Site-Level Result

Field	Description
<code>data[].systemId</code>	The unique site identifier for this result item.
<code>data[].result</code>	The processing result for the site. Use it together with <code>codeList</code> to determine the business status.
<code>data[].codeList</code>	The result code list for the site. Use this field to identify success, pending owner approval, missing permission, or the failure reason.

Important notes:

For onboard, `codeList=[1]` means the owner approval email was sent. The onboard request has been created, but the site is not finally onboarded until the owner approves it.

For offboard, `codeList=[1401]` means the current app has no permission to operate this site. The platform does not revoke the authorization, and callers should not treat it as a successful offboard.

4G sites are currently excluded from onboard processing and may not appear in the onboard data result list.

Onboard Result Codes

Code	Message	Reason	Suggested Action
0	Success	The site was onboarded successfully. If the site had already been onboarded by the current app, success is also returned.	Continue with APIs that require onboard permission.
1	Email Sent Successfully	Owner approval is required for the current developer, and the platform has sent the confirmation email to the site owner.	Show the status as pending owner approval. Onboard becomes effective only after the owner approves it.
1000	Param illegal	The request parameters are invalid, for example an empty request body, an empty systemId, or incomplete owner approval request parameters.	Check the request body format and systemId, then submit again.
1103	In-other-vpp	The site has already been onboarded by another VPP, service provider, or customer. The current app cannot onboard it again.	Check the current authorization relationship. To switch providers, the current onboarded party should offboard the site first.
1106	Station was not found	The platform could not find the site when creating the owner approval request.	Verify that the systemId is correct, or ask the site owner to check the site status.
1113	Station owner email missing	The platform could not find a valid site owner email, so the owner approval email cannot be sent.	Ask the site owner to complete or bind a valid email address, then retry onboard.

Code	Message	Reason	Suggested Action
1115	Notification failed	The owner approval email failed to send.	Retry later. If the issue persists, contact Sigenenergy support.
1116	Onboard request already pending	The current app already has a valid pending onboard request for this site.	Do not submit repeatedly. Wait for the owner to process the current email, or retry after the pending request expires.
1302	Station status anomaly	The site is in an abnormal status and cannot be onboarded at this time.	Retry after the site returns to normal, or ask the site owner to check the site status.
1502	Sigenenergy system internal error.	Internal platform processing failed, for example operation mode switching, authorization record writing, or another system error.	Retry later. If it occurs repeatedly, contact Sigenenergy support with the systemId and request time.

Offboard Result Codes

Code	Message	Reason	Suggested Action
0	Success	The site was offboarded successfully. If the current app has no active onboard relationship with the site, it is also treated as idempotent success.	Stop controlling, subscribing to, or querying data from this site.
1401	No permission to operate this station.	The current app is not the onboarded party for this site and cannot revoke its authorization.	Check whether the current app is the onboarded party. If the site needs to be released,

Code	Message	Reason	Suggested Action
			the current onboarded party should call offboard.
1502	Sigenergy system internal error.	Internal platform processing failed, for example exiting VPP mode or deleting the authorization record failed.	Retry later. If it occurs repeatedly, contact Sigenergy support with the <code>systemId</code> and request time.

Request-Level Errors

The following errors are usually returned in the top-level code. In these cases, `data[].codeList` may not be returned:

Code	Message	Reason	Suggested Action
1000	param illegal	The request body is empty, the format is invalid, or <code>systemIdList</code> is empty.	Check that the request body is a JSON array and includes at least one valid <code>systemId</code> .
11000	User info empty	The platform could not obtain the current app or account information.	Check whether the access token and app credentials are correct and valid.
13001	station info not found	None of the requested sites were found, so the platform cannot build site-level result items.	Check whether the <code>systemId</code> values are correct and whether the sites exist and can be recognized by the OpenAPI platform.

This chapter is mainly used to obtain information about authorized power stations and equipment. For service providers with keys, they mainly obtain data of power stations that have undergone the onboard process but not the offboard process. For service providers using accounts, the data of power stations within the account's authority are obtained.

It should be noted that these two processes are isolated from each other. If a service provider has both of these authorization methods, it cannot cross-access the power station data under the corresponding authorities.

Used to query the list of power stations that meet the conditions, supporting filtering by grid connection time.

Access Restriction

An account can only access every five minutes.

If only the grid connection start time is passed in, the grid connection end time will default to the current time. If only the grid connection end time is passed in, the grid connection start time will default to 08:00:00 on January 1, 1970.

Request Parameter

Name	Type	Required	Description
startTime	Long	No	Device grid connect start time, timestamp in seconds.
endTime	Long	No	Device grid connect end time, timestamp in seconds.

Response Message

Name	Type	Description
code	Integer	Error code
msg	String	Message
data	Object	The data contains the returned information, including the following:
> systemId	String	System id.

Name	Type	Description
> systemName	String	System name
> addr	String	System address
> status	String	System operational status
> isActivate	Boolean	System activated
> onOffGridStatus	String	Grid-Connected and off-Grid Status
> timeZone	String	System time zone
> gridConnectTime	Long	Grid Connection Time
> pvCapacity	Integer	PV capacity
> batteryCapacity	Integer	Battery capacity

Sample Request

```
{
  "startTime" : "1711468800",
  "endTime" : "1711814399"
}
```

Sample Successful Response

```
{
  "code": 0,
  "msg": "success",
  "data": [{
    "systemId": "NDXZZ1731665796",
    "systemName": "pfh24",
    "addr": "*** Shanghai China",
    "status": "normal",
  ]
}
```

```
        "isActive": true,  
        "onOffGridStatus": "onGrid",  
        "timeZone": "Asia/Shanghai",  
        "gridConnectTime": 1711814399,  
        "pvCapacity": 80,  
        "batteryCapacity": 8  
    }  
}
```

Sample Failed Response

```
{  
    "code": 1000,  
    "msg": "param illegal",  
    "data": null  
}
```

Used to query a list of stations that meet the specified criteria. Supports pagination and filtering by grid-connection time.

Access Restrictions

Each account can **access only once every 5 minutes**.

Each page can return a maximum of **100 station records**.

If only the grid connection start time (`startTime`) is provided, the grid connection end time (`endTime`) defaults to the current time.

If only the grid connection end time (`endTime`) is provided, the grid connection start time (`startTime`) defaults to **January 1, 1970, 08:00:00**.

Request Parameters

Parameter	Type	Required	Description
<code>startTime</code>	Long	No	Station grid connection start time, in seconds (timestamp)
<code>endTime</code>	Long	No	Station grid connection end time, in seconds (timestamp)
<code>pageNum</code>	Integer	Yes	Page number for paginated queries
<code>pageSize</code>	Integer	No	Number of records per page, defaults to 100

Response Parameters

Parameter	Type	Description
<code>code</code>	Integer	Error code

Parameter	Type	Description
msg	String	Message
data	Object	Returned data object, containing the following fields:
> total	Long	Total number of stations
> size	Long	Records per page
> current	Long	Current page number
> pages	Long	Total number of pages
> records	List	List of station data for the current page, containing the following fields:
>> systemId	String	System ID
>> systemName	String	System name
>> addr	String	System address
>> status	String	System operating status
>> isActivate	Boolean	Whether the system is activated
>> onOffGridStatus	String	On-grid / Off-grid status
>> timeZone	String	System time zone
>> gridConnectTime	Long	Grid connection time (timestamp)
>> pvCapacity	Integer	PV capacity
>> batteryCapacity	Integer	Battery capacity

Request Example

```
{
  "startTime" : "1711468800",
  "endTime" : "1711814399",
  "pageNum" : 0,
  "pageSize" : 10
}
```

Success Response Example

```
{
  "code": 0,
  "msg": "success",
  "timestamp": 1761189111,
  "data": {
    "records": [
      {
        "\systemId\":"IXXMQ1743989898\","systemName\":"L1\","addr\":"Minhang District, Shanghai, China\","status\":"Normal\","isActivate\":true,"onOffGridStatus\":"on Grid\","timeZone\":"Asia/Shanghai\","gridConnectedTime\":1743665483000, "\pvCapacity\":5, "\batteryCapacity\":5}
      },
      {
        "\systemId\":"SGHG01761898989\","systemName\":"L2\","addr\":"Minhang District, Shanghai, China\","status\":"Disconnection\","isActivate\":true,"onOffGridStatus \":"onGrid\","timeZone\":"Asia/Shanghai\","gridConnectedTime\":1761101752000, "\pvCapacity\":50, "\batteryCapacity\":50}
      }
    ],
    "total": 2,
    "size": 10,
    "current": 1,
    "pages": 1
  }
}
```

Failure Response Example

```
{  
  "code": 1000,  
  "msg": "param illegal",  
  "data": null  
}
```

Query all device information under the power station based on the unique power station identifier (systemId).

Access Restriction

One account can only access one station's device list once every five minutes.

Request Parameter

Name	Type	Required	Description
systemId	String	Yes	System unique code

Response Message

Name	Type	Description
code	Integer	Error Code
msg	String	Message
data	List	Return data, The data contains the returned information, including the following:
> systemId	String	System unique code
> serialNumber	String	Device serial number
> deviceType	String	Type of device. Refer to Enum Data
> status	String	Device status. Refer to Enum Data
> pn	String	Different PN codes identify different types of system.

Name	Type	Description
> firmwareVersion	String	Firmware version.
> attrMap	Object	Attribute Object. Refer to the following attribute information for each device:

Inverter Attribute

Key	Description	Unit	Data Type
ratedActivePower	Designated power output capacity.	kW	Double
maxActivePower	Maximum achievable power output.	kW	Double
maxAbsorbedPower	Highest power absorption capacity.	kW	Double
ratedVoltage	Specified operating voltage.	V	Double
ratedFrequency	Prescribed operating frequency.	Hz	Double
pvStringNumber	Number of PV strings in the system.	-	Integer

Battery Attribute

Key	Description	Unit	Data Type
ratedEnergy	Rated Energy Storage Capacity	kWh	Double
chargeableEnergy	Maximum Rechargeable Energy	kWh	Double
dischargeEnergy	Maximum Dischargeable Energy	kWh	Double
ratedChargePower	Rated Charging Power	kW	Double
ratedDischargePower	Rated Discharging Power	kW	Double

Sample Request

```
{  
  "systemId" : "NDXZZ1731665796"  
}
```

Sample Successful Response

```
{  
  "code": 0,  
  "msg": "success",  
  "data": [{  
    "systemId": "NDXZZ1731665796",  
    "serialNumber": "110A118B0028",  
    "deviceType": "Inverter",  
    "status": "normal",  
    "pn": "1711814399",  
    "firmwareVersion": "V100R001C22B028",  
    "attrMap" : {  
      "ratedActivePower": 8.0,  
      "maxActivePower": 8.0,  
      "maxAbsorbedPower": 8.0,  
      "ratedVoltage": 8.0,  
      "ratedFrequency": 8.0,  
      "pvStringNumber": 8.0  
    }  
  }]  
}
```

Sample Failed Response

```
{  
  "code": 1000,  
  "msg": "param illegal",  
  "data": null  
}
```

System Status Enumeration

status	Description
Standby	No power is activated.
Normal	All devices in the system are operating normally.
Fault	At least one device in the system is malfunctioning.
Offline	Device communication with the cloud is interrupted.

System Type Enumeration

type	Description
Residential	Residential solar system.
Commercial	Commercial and industrial rooftop system.

Network Connection Type Enumeration

type	Description
WIFI	Wireless network for local/internet access.
4G	Mobile broadband internet connection.
FE	Wired connection via fiber Ethernet for high-speed data transfer.

Device Type Enumeration

Device type	Description
Inverter	Converts DC from solar panels into AC for home use and grid connection.
Battery	Energy storage system that stores excess solar energy for use when sunlight is insufficient.
Gateway	Central communication device that connects PV system components to the monitoring platform for remote control and analysis.
DcCharger	Charges batteries directly using PV DC power, optimizing the charging process.
AcCharger	Charges electric vehicles or storage devices using grid AC or PV-converted AC.
Meter	Measures PV system generation, consumption, and grid feed-in.

Inverter Status Enumeration

Device Status	Description
Standby	No power is activated.
Normal	Inverter is operating normally.
Fault	At least one device in the system is malfunctioning.
Shutdown	Inverter is shut down.
Offline	Inverter is offline (communication interrupted).

Battery Status Enumeration

Device Status	Description
Standby	Battery is in standby mode.
Normal	Battery is charging or discharging.
Fault	Battery system fault.
Dormancy	Battery is not activated.
Offline	Battery is offline (communication interrupted).

DC Charger Status Enumeration

Device Status	Description
Init	No power is activated.
Idle	DC charger is idle.
Normal	All devices in the system are operating normally.
Fault	At least one device in the system is malfunctioning.
Shutdown	DC charger is shut down.
Reset	DC charger is resetting.
EmergencyStopped	DC charger is in emergency stop.
Offline	DC charger is offline (communication interrupted).

AC Charger Status Enumeration

Device Status	Description
IdleUnplugged	AC charger is idle (plug not inserted).
OccupiedNotStarted	AC charger is occupied but not charging.
PreparingWaitingCarStart	AC charger is ready, waiting for vehicle start signal.
Charging	AC charger is charging.
Fault	AC charger fault.
Scheduled	AC charger is scheduled.
Offline	AC charger is offline (communication interrupted).

Gateway Status Enumeration

Device Status	Description
Normal	Gateway is operating normally.
Fault	Gateway fault.
Offline	Gateway is offline (communication interrupted).

Meter Status Enumeration

Device Status	Description
Normal	Meter is operating normally.
Offline	Meter is offline (communication interrupted).

System Historical Data

Type	Description
Request Method	GET
Access Restriction	One account can only access one station once every five minutes.

Request Parameter

Name	Type	Required	Description
level	String	Yes	History statistics data level. Enum Data
date	String	Optional	Date, format: yyyy-MM-dd. Day, Month, Year is required, and Lifetime is not required.

Response Message

Name	Type	Description
code	Integer	Error Code
msg	String	Message
data	Object	The data contains the returned information, including the following.
>powerGeneration	Double	Total green power generation.
>powerToGrid	Double	Green power sold to the grid.

Name	Type	Description
>powerSelfConsumption	Double	Self-consumed green power.
>powerUse	Double	Total load power consumption.
>powerFromGrid	Double	Non-green power purchased from the grid.
>powerOneself	Double	Load green power consumption.
>esCharging	Double	Total battery charging energy.
>esDischarging	Double	Total battery discharging energy.
>itemList	List	Data at Each Time Point including the following.
>>dateTime	String	Data timestamp.
>>pvTotalPower	Double	Solar real-time power output.
>>loadPower	Double	Household load real-time power draw.
>>toGridPower	Double	Realtime power exported to the grid.
>>fromGridPower	Double	Realtime power imported from the grid.
>>esChargeDischargePower	Double	Battery charge/discharge real-time power.
>>esChargePower	Double	Battery charging real-time power.
>>esDischargePower	Double	Battery discharging real-time power.
>>oneselfPower	Double	Self-sufficiency real-time power usage.
>>powerGeneration	Double	Total green power generation.
>>powerToGrid	Double	Green power sold to the grid.

Name	Type	Description
>>powerSelfConsumption	Double	Self-consumed green power.
>>powerUse	Double	Total load power consumption.
>>powerFromGrid	Double	Non-green power purchased from the grid.
>>powerOneself	Double	Load green power consumption.
>>powerFromBattery	Double	Power drawn from the battery.
>>esCharging	Double	Battery charging energy.
>>esDischarging	Double	Battery discharging energy.
>>batSoc	Double	Battery State of Charge (available on a day-level basis only).

Sample Request

```
"systemId" : "NDXZZ1731665796"
{
  "level" : "Day",
  "date" : "2024-06-27"
}
```

Sample Successful Response

```
{
  "code": 0,
  "msg": "success",
  "data": {
    "powerGeneration": 22.94,
    "powerToGrid": 3.93,
    "powerSelfConsumption": 19.01,
    "powerUse": 24.83,
```



```
"powerFromGrid": 9.67,  
"powerOneself": 15.16,  
"esCharging": 6.51,  
"esDischarging": 0.47,  
"itemList": [{  
  "dateTime": "2024-04-02 00:00",  
  "pvTotalPower": 0.0,  
  "loadPower": 1.06,  
  "toGridPower": 0.0,  
  "fromGridPower": 0.0,  
  "esChargeDischargePower": -1.016,  
  "esChargePower": 0.0,  
  "esDischargePower": 1.016,  
  "oneselfPower": 1.06,  
  "powerGeneration": 0.0,  
  "powerToGrid": 0.0,  
  "powerSelfConsumption": 0.0,  
  "powerUse": 0.01,  
  "powerFromGrid": 0.0,  
  "powerOneself": 0.01,  
  "powerFromBattery": 0.0,  
  "esCharging": 0.0,  
  "esDischarging": 0.0,  
  "batSoc": 20.1  
}]  
}
```

Sample Failed Response

```
{  
  "code": 1000,  
  "msg": "param illegal",  
  "data": null  
}
```

Device Historical Data

Type	Description
Request Method	GET
Access Restriction	One account can only access one device in a station once every five minutes.

Request Parameter

Name	Type	Required	Description
level	String	Yes	History statistics data level. Enum Data
date	String	Optional	Date, format: yyyy-MM-dd. Day, Month, Year is required, and Lifetime is not required.

Response Message

Name	Type	Description
code	Integer	Error Code
msg	String	Message
data	Object	The data contains the returned information, including the following.
>systemId	String	System unique code

Name	Type	Description
>serialNumber	String	Device serial number
>deviceType	String	Device type
>itemList	List	Data at Each Time Point including the following.

Inverter History Data

Key	Description	Unit	Data Type
activePower	Real-time active power output	kW	Numeric
reactivePower	Real-time reactive power	kW	Numeric
aPhaseVoltage	Voltage of phase A	V	Numeric
bPhaseVoltage	Voltage of phase B	V	Numeric
cPhaseVoltage	Voltage of phase C	V	Numeric
aPhaseCurrent	Current flowing through phase A	A	Numeric
bPhaseCurrent	Current flowing through phase B	A	Numeric
cPhaseCurrent	Current flowing through phase C	A	Numeric
powerFactor	Ratio of active power to apparent power	-	Numeric
gridFrequency	Frequency of the electric grid	Hz	Numeric
pVPower	Total power produced by solar panels	kW	Numeric
pV1Voltage	Voltage of PV string 1	V	Numeric
pV1Current	Current from PV string 1	A	Numeric
pV2Voltage	Voltage of PV string 2	V	Numeric

Key	Description	Unit	Data Type
pV2Current	Current from PV string 2	A	Numeric
pV3Voltage	Voltage of PV string 3	V	Numeric
pV3Current	Current from PV string 3	A	Numeric
pV4Voltage	Voltage of PV string 4	V	Numeric
pV4Current	Current from PV string 4	A	Numeric

Battery History Data

Key	Description	Unit	Data Type
batterySOC	State of Charge of the Battery	%	Numeric
chargingDischargingPower	Power being consumed or supplied to the battery	kW	Numeric
chargeEnergy	Battery charging energy.	kWh	Numeric
dischargeEnergy	Battery discharging energy.	kWh	Numeric

Sample Request

```
"systemId" : "NDXZZ1731665796"
"serialNumber" : "110A118B0028s"

{
  "level" : "Day",
  "date" : "2024-06-27"
}
```

Sample Successful Response

```
{
  "code": 0,
  "msg": "success",
  "data": {
    "systemId": "NDXZZ1731665796",
    "serialNumber": "110A118B0028",
    "deviceType": "Inverter",
    "itemList": [{
      "dateTime": "2024-04-02 00:00"
    }]
  }
}
```

Sample Failed Response

```
{
  "code": 1000,
  "msg": "param illegal",
  "data": null
}
```

Request Parameters

Field	Type	Required	Description
systemId	String	Yes	Unique system identifier
date	String	Yes	Query date in the format: YYYY-MM-DD
level	String	Yes	Time dimension (Day = daily, Week = weekly, Month = monthly, Year = yearly, Lifetime = full lifecycle)

Request Example

```
"systemId": "NDXZZ1731665796",  
"date": "2025-08-06",  
"level": "Day"
```

Response Example

```
{  
  "code": 0,  
  "msg": "success",  
  "data": {  
    "sankeyData": {  
      "nodes": [  
        {  
          "id": "FROM_SOLAR",  
          "value": 66.03  
        },  
        {  
          "id": "FROM_BATTERY",  
          "value": 0  
        }  
      ],  
      "links": []  
    }  
  }  
}
```

```
{
  "id": "TO_BATTERY",
  "value": 3.88
},
{
  "id": "TO_EVDC",
  "value": 0
},
{
  "id": "FROM_EVDC",
  "value": 0.11
},
{
  "id": "FROM_GRID",
  "value": 1.11
},
{
  "id": "TO_LOAD",
  "value": 1.21
},
{
  "id": "TO_GRID",
  "value": 62.16
}
],
"links": [
  {
    "sourceId": "FROM_SOLAR",
    "targetId": "TO_BATTERY",
    "value": 3.87
  },
  {
    "sourceId": "FROM_SOLAR",
    "targetId": "TO_GRID",
    "value": 62.16
  },
  {
    "sourceId": "FROM_EVDC",
    "targetId": "TO_BATTERY",
    "value": 0.01
  },
  {
    "sourceId": "FROM_EVDC",
    "targetId": "TO_LOAD",
    "value": 0.1
  },
  {
```

```
        "sourceId": "FROM_GRID",
        "targetId": "TO_LOAD",
        "value": 1.11
    }
]
},
"chartData": {
    "chartName": "Energy",
    "unit": "kW",
    "chartType": "LINE",
    "dataSeries": [
        {
            "id": "FROM_SOLAR",
            "points": [
                {
                    "time": "20250806 00:00",
                    "value": 4.79
                },
                {
                    "time": "20250806 14:20",
                    "value": 4.69
                }
            ]
        },
        {
            "id": "BATTERY",
            "points": [
                {
                    "time": "20250806 00:00",
                    "value": 0
                },
                {
                    "time": "20250806 14:20",
                    "value": -0.9
                }
            ]
        },
        {
            "id": "TO_EVDC",
            "points": [
                {
                    "time": "20250806 00:00",
                    "value": 0
                },
                {
                    "time": "20250806 14:20",
                    "value": 0
                }
            ]
        }
    ]
}
```



```
    }
  ]
},
{
  "id": "FROM_EVDC",
  "points": [
    {
      "time": "20250806 00:00",
      "value": 0
    },
    {
      "time": "20250806 14:20",
      "value": 0
    }
  ]
},
{
  "id": "GRID",
  "points": [
    {
      "time": "20250806 00:00",
      "value": -4.79
    },
    {
      "time": "20250806 14:20",
      "value": -3.79
    }
  ]
},
{
  "id": "TO_LOAD",
  "points": [
    {
      "time": "20250806 00:00",
      "value": 0
    },
    {
      "time": "20250806 14:20",
      "value": -0.01
    }
  ]
}
],
"preTimePeriodTotal": null
}
}
```

Data Interpretation

Sankey Diagram Usage

The Sankey diagram visualizes energy flow through the power station system:

Nodes represent energy sources, storage, and consumption points

Links show the flow of energy between components

Values indicate the amount of energy transferred

Chart Data Usage

The chart data provides time-series information for detailed analysis:

Positive values typically represent energy generation or import

Negative values typically represent energy consumption or export

Time points are provided at regular intervals based on the selected time dimension

Value Interpretation

FROM_SOLAR: Solar energy generation (always positive)

BATTERY: Positive = discharging, Negative = charging

GRID: Positive = importing from grid, Negative = exporting to grid

TO_LOAD: Energy consumption by loads (usually positive or zero)

TO_EVDC/FROM_EVDC: EV charger energy flow

Energy Component IDs

Source Components (Energy Generation)

FROM_SOLAR – Solar panel energy generation

FROM_BATTERY - Energy discharged from battery

FROM_EVDC - Energy from EV DC charger

FROM_GRID - Energy imported from grid

Target Components (Energy Consumption)

T0_BATTERY - Energy charged to battery

T0_EVDC - Energy sent to EV DC charger

T0_LOAD - Energy consumed by loads

T0_GRID - Energy exported to grid

Bidirectional Components

BATTERY - Battery charge/discharge (positive = discharge, negative = charge)

GRID - Grid import/export (positive = import, negative = export)

Response Structure

C0

Field	Type	Description
sankeyData	C1	Sankey diagram data for visualizing cumulative energy flows within the selected time frame
chartData	C2	Time-series energy chart data, historical values at each sampling point

C1: Sankey Data

Field	Type	Description
nodes	List<C1-I>	Node list representing energy system components

Field	Type	Description
links	List<C1-2>	Link list representing energy flows between nodes

C1-1: Node

Field	Type	Description
id	String	Unique identifier of the energy system component
value	Double	Total energy associated with this component

C1-2: Link

Field	Type	Description
sourceId	String	Identifier of the source node
targetId	String	Identifier of the target node
value	Double	Energy flow amount between the nodes

C2: Chart Data

Field	Type	Description
chartName	String	Chart display name
unit	String	Data unit (e.g., kW, kWh)
chartType	String	Chart type (e.g., LINE, BAR)
dataSeries	List<C2-1>	Collection of data series
preTimePeriodTotal	Double	Total from the previous period (nullable)

C2-1: Data Series

Field	Type	Description
id	String	Data series identifier
points	List<C3>	Collection of data points

C3: Data Point

Field	Type	Description
time	String	Timestamp in YYYYMMDD HH:mm format
value	Double	Energy value at the given timestamp

Notes

Data Consistency:

- Sankey data shows **cumulative totals** for the selected period
- Chart data provides **time-series values**

Energy Balance:

Total energy input $\approx$ total output + storage variation

Data Level Enum

Level	Description
Day	Displays data at a 5-minute granularity within one day.
Week	Present the data of the week in which the target date falls.
Month	Presents daily data for an entire month.
Year	Shows monthly data across a whole year.
Lifetime	Illustrates annual data throughout the system's or product's lifetime.

Used to obtain real-time summary data of the specified power station system, including daily, monthly, annual and cumulative power generation, as well as cumulative emission reduction related data.

Access Restriction

One account can only access one station once every five minutes.

Request Parameter

Name	Type	Required	Description
systemId	String	Yes	System unique code

Response Message

Name	Type	Description
dailyPowerGeneration	Double	Daily PV energy generation (kWh)
monthlyPowerGeneration	Double	Monthly PV energy generation (kWh)
annualPowerGeneration	Double	Annual PV energy generation (kWh)
lifetimePowerGeneration	Double	Lifetime PV energy generation (kWh)
lifetimeCo2	Double	Lifetime CO ₂ emission reduction (tons)
lifetimeCoal	Double	Lifetime standard coal saved (tons)
lifetimeTreeEquivalent	Double	Lifetime equivalent number of trees planted

Sample Request

```
{  
  "systemId" : "NDXZZ1731665796"  
}
```

Sample Successful Response

```
{  
  "code": 0,  
  "msg": "success",  
  "timestamp": 1757581478,  
  "data": {  
    "dailyPowerGeneration": 0.0,  
    "monthlyPowerGeneration": 0.0,  
    "annualPowerGeneration": 1394.37,  
    "lifetimePowerGeneration": 1394.38,  
    "lifetimeCo2": 0.66,  
    "lifetimeCoal": 0.56,  
    "lifetimeTreeEquivalent": 0.9  
  }  
}
```

Sample Failed Response

```
{  
  "code": 1000,  
  "msg": "param illegal",  
  "data": null  
}
```


Used to obtain real-time energy flow data of the specified power station, including power of PV, grid, load, energy storage and other links, and energy storage SOC status.

Access Restriction

One account can only access one device in a station once every five minutes.

Request Parameter

Name	Type	Required	Description
systemId	String	Yes	System unique code

Response Message

Name	Type	Description
code	Boolean	Error Code (0 means success)
msg	String	Message
data	Object	The data contains the returned information, including the following:
>> pvPower	Double	PV generation power (unit: kW)
>> gridPower	Double	Grid power (positive is selling to grid and negative is buying from grid, unit: kW)
>> evPower	Double	DC or AC power (unit: kW)
>> loadPower	Double	Used by load (unit: kW)

Name	Type	Description
>> heatPumpPower	Double	Heat pump power (unit: kW)
>> batteryPower	Double	Battery power (positive is import to batteries, and negative is export from battery, unit: kW)
>> batterySoc	Double	Battery soc (unit: %)

Sample Request

```
{  
  "systemId" : "NDXZZ1731665796"  
}
```

Sample Successful Response

```
{  
  "code": 0,  
  "msg": "success",  
  "data": {  
    "pvPower": 10.1,  
    "gridPower": 10.1,  
    "evPower": 0,  
    "loadPower": 0,  
    "heatPumpPower": 0,  
    "batteryPower": 0,  
    "batterySoc": 100  
  }  
}
```

Sample Failed Response

```
{  
  "code": 1000,
```

```
"msg": "param illegal",  
"data": null  
}
```

This interface is used to obtain real-time operation data of a specified device under a specified power station, supporting real-time information query for device types such as AIO, batteries, gateways, and meters.

Access Restriction

One account can only access one device in a station once every five minutes.

Request Parameters

Parameter Name	Type	Required	Description
systemId	String	Yes	Unique identifier of the power station
serialNumber	String	Yes	Device serial number

Response Parameters

Parameter Name	Type	Description
code	Boolean	Error code
msg	String	Prompt message
data	Object	Returned data, including the following content
> systemId	String	Unique identifier of the power station
> serialNumber	String	Device serial number
> deviceType	String	Device type

Parameter Name	Type	Description
> realTimeInfo	Object	Device real-time information (different fields for different device types)

Real-time Data Fields by Device Type

AIO Realtime Data

Key	Description	Unit	Data Type
activePower	Real-time active power output (positive: generation, negative: consumption)	kW	Numeric
reactivePower	Real-time reactive power (positive: generation, negative: consumption)	kW	Numeric
aPhaseVoltage	Voltage of phase A	V	Numeric
bPhaseVoltage	Voltage of phase B	V	Numeric
cPhaseVoltage	Voltage of phase C	V	Numeric
aPhaseCurrent	Current of phase A	A	Numeric
bPhaseCurrent	Current of phase B	A	Numeric
cPhaseCurrent	Current of phase C	A	Numeric
powerFactor	Power factor ($\cos\phi$)	-	Numeric
gridFrequency	Grid frequency	Hz	Numeric
pvPower	Total PV power currently generated	kW	Numeric

Key	Description	Unit	Data Type
pv1Voltage	PV string 1 voltage	V	Numeric
pv1Current	PV string 1 current	A	Numeric
pv2Voltage	PV string 2 voltage	V	Numeric
pv2Current	PV string 2 current	A	Numeric
pv3Voltage	PV string 3 voltage	V	Numeric
pv3Current	PV string 3 current	A	Numeric
pv4Voltage	PV string 4 voltage	V	Numeric
pv4Current	PV string 4 current	A	Numeric
internalTemperature	Device internal temperature	°C	Numeric
insulationResistance	Device insulation resistance	MΩ	Numeric
pvEnergyDaily	Daily PV energy generation	kWh	Numeric/String
pvEnergyTotal	Total PV energy generation (lifetime)	kWh	Numeric/String
batPower	Battery real-time power (positive: discharging, negative: charging)	kW	Numeric
pvTotalPower	Combined PV output power	kW	Numeric
pcsReactivePower	PCS (Power Conversion System) reactive power	kW	Numeric
pcsActivePower	PCS (Power Conversion System) active power	kW	Numeric
esDischargingDay	Battery energy discharged today	kWh	Numeric
esChargingDay	Battery energy charged today	kWh	Numeric

Key	Description	Unit	Data Type
pvPowerDay	Daily PV power	kWh	Numeric
esDischargingTotal	Total battery discharged energy (lifetime)	kWh	Numeric
batSoc	Battery State of Charge	%	Numeric

Gateway Realtime Data

Key	Description	Unit	Data Type
voltageA	Phase A voltage	V	Numeric
voltageB	Phase B voltage	V	Numeric
voltageC	Phase C voltage	V	Numeric
currentA	Phase A current	A	Numeric
currentB	Phase B current	A	Numeric
currentC	Phase C current	A	Numeric
activePower	Phase A apparent power	kW	Numeric
reactivePower	Phase A reactive power	kW	Numeric

Meter Realtime Data

Key	Description	Unit	Data Type
voltageA	Phase A voltage	V	Numeric
voltageB	Phase B voltage	V	Numeric

Key	Description	Unit	Data Type
voltageC	Phase C voltage	V	Numeric
currentA	Phase A current	A	Numeric
currentB	Phase B current	A	Numeric
currentC	Phase C current	A	Numeric
powerFactor	Power factor	-	Numeric
gridFrequency	Grid frequency	Hz	Numeric
activePower	Total active power (consumption/generation)	kW	Numeric
reactivePower	Total reactive power (consumption/generation)	kW	Numeric

Sample Request

```
{  
  "systemId" : "NDXZZ1731665796",  
  "serialNumber" : "110B115K0053"  
}
```

Sample Successful Response

```
{  
  "code": 0,  
  "msg": "success",  
  "timestamp": 1757583276,  
  "data": {  
    "systemId": "NDXZZ1731665796",  
    "serialNumber": "110B115K0053",  
    "deviceType": "Inverter",  
  }  
}
```



```
"realTimeInfo": {
  "activePower": -0.115,
  "reactivePower": -0.005,
  "aPhaseVoltage": 230.75,
  "bPhaseVoltage": 235.62,
  "cPhaseVoltage": 233.66,
  "aPhaseCurrent": 0.76,
  "bPhaseCurrent": 0.81,
  "cPhaseCurrent": 0.73,
  "powerFactor": -0.996,
  "gridFrequency": 49.97,
  "pvPower": 0.0,
  "pv1Voltage": 0.0,
  "pv1Current": 0.0,
  "pv2Voltage": 0.0,
  "pv2Current": 0.0,
  "pv3Voltage": 0.0,
  "pv3Current": 0.0,
  "internalTemperature": 54.7,
  "insulationResistance": 0.685,
  "pvEnergyDaily": "0.00",
  "pvEnergyTotal": "1299.16",
  "pv4Voltage": -0.1,
  "pv4Current": -0.01,
  "batPower": -0.001,
  "pvTotalPower": 0.0,
  "pcsReactivePower": -0.005,
  "pcsActivePower": -0.115,
  "esDischargingDay": 0.01,
  "esChargingDay": 0.0,
  "pvPowerDay": 0.0,
  "esDischargingTotal": 1077.98,
  "batSoc": 39.0
}
```

Sample Failed Response

```
{
  "code": 1000,
  "msg": "param illegal",
}
```

```
"data": null  
}
```

MQTT is a message transmission protocol based on the publish/subscribe model, specifically designed for constrained devices and networks with low bandwidth, high latency, or unreliability. It has the following remarkable advantages:

Lightweight: The protocol has minimal header overhead, reducing network bandwidth consumption. It is suitable for use in devices and network environments with limited resources.

Real-time: It supports real-time message pushing, ensuring that data can be transmitted promptly. This meets the system's real-time requirements, whether it is regular data or urgent alarm information.

Reliability: It offers three different levels of Quality of Service (QoS). Users can select an appropriate QoS level according to actual needs to ensure the reliable transmission of messages. For important alarm information, a high - reliability transmission level can be chosen.

Flexibility: Adopting the publish/subscribe model, it allows for loosely - coupled communication between devices. This improves the system's scalability and flexibility, facilitating the classification management and subscription of different types of messages, including alarm information.

Our service supports actively pushing system data to a specified server, and it also covers the pushing of device alarm information. Before pushing data, the following operations are required:

Connect to the Broker Server The Broker server is a core component of the MQTT protocol, responsible for receiving, storing, and forwarding messages. Before pushing data, you need to connect to our Broker server. The connection process is as follows:

Obtain connection information: We will provide the address (such as an IP address or domain name) and port number of the Broker server.

Establish a connection: Use an MQTT client library to establish a connection to the Broker server based on the obtained connection information. During the connection process, you may need to provide authentication information such as a client ID, username, and password to ensure the security of the connection.

Configure Topic Information A Topic is a string in the MQTT protocol used to identify the message subject, similar to a message classification label. When publishing and subscribing to messages, you need to specify the corresponding Topic. Our data pushing service divides Topics into three categories according to data types:

Real-time operation data Topic: Used to push real-time operation data such as voltage, current, power, and frequency. The system will push this data regularly according to a set cycle.

Rated parameter and system status data Topic: Used to push rated parameter and system status data. The system will only push this data when there are changes.

Alarm information Topic: Specifically used to push device alarm information. Once a device detects an abnormal situation, such as high temperature or abnormal pressure, the relevant alarm information will be immediately pushed to this Topic. The alarm information includes key content such as the alarm type, occurrence time, and device identification, which is convenient for the recipient to quickly locate and handle the problem. When configuring Topic information, you need to clarify the name and purpose of each Topic and subscribe to the corresponding Topics as needed.

Service providers have the ability to send instructions to specified topics via MQTT. When our system receives data from a specific topic, it will automatically execute the corresponding functions based on the pre - configured interfaces. For example, service providers can send instructions to adjust device operation parameters through a specific topic. After our system receives such an instruction, it will accurately call the corresponding interface to complete the adjustment of device parameters. It should be noted that during the data transmission process, the size limit for a single piece of data is 256KB. This specification ensures the stable and efficient transmission of data across different devices and network environments, avoiding transmission errors or performance issues caused by overly large data. Strictly adhering to the MQTT data size standard helps maintain the stability and reliability of the entire message transmission system and ensures the smooth progress of the instruction sending and execution process.

Periodic Pushing For real-time operation data, the system will push the data regularly according to a set cycle. For example, it will push data such as voltage, current, power, and frequency every minute. The recipient can adjust the subscribed Topics and processing logic as needed to meet actual requirements.

Change Pushing For rated parameter and system status data, the system will only push the data when there are changes. This can reduce unnecessary data transmission and improve system efficiency. The recipient can promptly obtain information on data changes and make corresponding adjustments.

Alarm Pushing For alarm information, once a device triggers an alarm, the relevant information will be immediately pushed to the alarm information Topic through the MQTT protocol. This real-time pushing mechanism ensures that the recipient can be informed of the device's abnormal conditions in a timely manner, so as to quickly take countermeasures and ensure the stable operation of the system.

To better receive different types of data, the recipient will need to subscribe to three different topics, which are respectively used to receive the periodically pushed real-time operation data, the change based pushed rated parameter and system status data, and the real-time pushed alarm information. This facilitates the recipient's classification, processing, and storage of different types of data. Corresponding processing strategies can be formulated according to the characteristics of different data, especially for urgent alarm information, which can be responded to quickly.

Through the above introduction, we believe you have gained a deeper understanding of our MQTT based data pushing service, including its support for alarm information. If you have any questions or need further assistance, please feel free to contact us.

For service providers who have a passive need to obtain power station data, we offer three data push methods: periodic push, push upon change, and alarm push.

Periodic Push: It mainly pushes data with a high frequency of change, such as electricity quantity, power, current, voltage, etc.

Change Push: It mainly pushes data with a low frequency of change, such as rated data, system configuration data, etc.

Alarm Push: When an alarm occurs in the power station, relevant alarm information will be pushed.

Service providers can obtain the data by executing the following subscription interface. If they no longer need the data, they can cancel the subscription. It should be noted that for the power stations where the interfaces in this chapter are executed, the service providers are required to have the execution authority for these power stations. If the service providers do not have the authority, customers with keys need to perform the onboard operation on the power stations.

Request Parameter

Name	Type	Required	Description
accessToken	String	Yes	The authorization code generated according to Chapter 2.
systemIdList	List	Yes	System unique code list

Sample Request

```
{  
  "accessToken": "4JT0b5E5WBtzM70PWiRZITKzN45URcq1",  
  "systemIdList": ["CBFGA1627168228"]  
}
```

Reminders

Data interaction is carried out in JSON format.

The api requires authorization in northbound scene.

Request Parameter

Name	Type	Required	Description
accessToken	String	Yes	The authorization code generated according to Chapter 2.
systemIdList	List	Yes	System unique code list

Sample Request

```
{
  "accessToken": "4JT0b5E5WBtzM70PWiRZITKzN45URcq1",
  "systemIdList": ["CBFGA1627168228"]
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Sample Request

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{
  "accessToken": "4JT0b5E5WBtzM70PWiRZITKzN45URcq1",
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}
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Reminders

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Request Parameter

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Sample Request

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{
  "accessToken": "4JT0b5E5WBtzM70PWiRZITKzN45URcq1",
  "systemIdList": ["CBFGA1627168228"]
}
```

Reminders

Data interaction is carried out in JSON format.

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Request Parameter

Name	Type	Required	Description
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systemIdList	List	Yes	System unique code list

Sample Request

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{
  "accessToken": "4JT0b5E5WBtzM70PWiRZITKzN45URcq1",
  "systemIdList": ["CBFGA1627168228"]
}
```

Reminders

Data interaction is carried out in JSON format.

The api requires authorization in northbound scene.

The Sigenergy Openapi is equipped with powerful and flexible data transmission capabilities. It can regularly and stably push the telemetry data of system equipment to the service according to a preset time cycle. This function plays a crucial role in ensuring the efficient and intelligent operation of the power system. Currently, to meet the diverse needs of different users and application scenarios, we especially support customized settings for data push intervals and content.

Users can determine the data push interval according to their actual business situations. In terms of data push content, a wide range of options are provided, covering the operating status data of equipment, such as equipment health status information, inverter power, grid connection point power, etc. This comprehensively helps users gain an in-depth understanding of the operating conditions of system equipment.

If you have any questions regarding the customization of data push intervals and content during the usage process, or if you need further technical support and detailed information, please do not hesitate to contact our professional support team. The members of our team all possess profound professional knowledge and rich practical experience. We will serve you wholeheartedly to ensure that you can make full use of all the functions of the Sigenergy Openapi and achieve the best system operation results.

Reminders

The push interval is 5 minutes at default.

The topics for telemetry data can be viewed under the "Data Subscription" section in the Control Center application of the Open Platform.

Data reception requires using the subscription interface in [Subscription For Telemetry](#).

If the pushed data does not meet your requirements, please contact technician.

Data Example

```
[
  {
    "deviceType": "system",
    "systemId": "AXBNH1123789222",
    "snCode": "AXBNH1123789222",
    "statisticsTime": 1757407020,
    "value": {
      "gridPhaseCReactivePowerVar": "-51.0",
      "inverterActivePowerW": "4207.0",
      "inverterReactivePowerVar": "-6.0",
      "inverterPhaseBReactivePowerVar": "0.0",
      "inverterMaxChargePowerW": "4199.0",
      "inverterMaxAbsorptionActivePowerW": "5000.0",
      "inverterPhaseAActivePowerW": "4192.0",
      "inverterPhaseBActivePowerW": "0.0",
      "gridActivePowerW": "-4151.0",
      "inverterMaxFeedInActivePowerW": "5000.0",
      "inverterMaxDischargePowerW": "2466.0",
      "inverterPhaseAReactivePowerVar": "0.0",
      "inverterPhaseCActivePowerW": "0.0",
      "inverterMaxFeedInReactivePowerVar": "3000.0",
      "storageChargeCapacityWh": "6090.0",
      "storageDischargeCapacityWh": "1960.0",
      "pvPowerW": "1740.0",
      "gridPhaseBActivePowerW": "0.0",
      "gridPhaseBReactivePowerVar": "0.0",
      "gridPhaseAActivePowerW": "-4184.0",
      "storageChargeDischargePowerW": "-2464.0",
      "storageSOC%": "24.4",
      "inverterPhaseCReactivePowerVar": "0.0",
      "gridPhaseCActivePowerW": "33.0",
      "gridReactivePowerVar": "-51.0",
      "inverterMaxAbsorptionReactivePowerVar": "3000.0",
      "gridPhaseAReactivePowerVar": "0.0"
    }
  }
]
```


Data Field Description

Name	Type	Description
systemId	String	The unique code representing the power station
snCode	String	Serial Number code of system
deviceType	String	Type of device: system
statisticsTime	Long	Time of statistics
value	String key-value	Telemetry data of interest

For detailed signal description, please refer to [Telemetry Signals Description](#)

In intelligent power systems, for certain information that does not change frequently, such as battery capacity and operational status, Sigenergy Openapi provides this data in full during the system onboarding process. This allows vendors to have a clear understanding of the system's basic configuration and initial operational status from the very beginning.

Once these data change subsequently, Sigenergy Openapi will immediately push the latest updated data to the vendors. This is crucial. For example, in battery charge and discharge operations, knowing the maximum charge and discharge power accurately is essential to properly set the power parameters, ensuring battery lifespan as well as system stability and safety.

Considering the diverse data needs of different users, Sigenergy Openapi supports customized data services. If you have any questions regarding data customization or the provision and update mechanism of the above information, and wish to learn more details, please contact our professional support team.

Notes

The push is triggered when the data changes.

The topic for system data can be viewed under the "Data Subscription" section in the Control Center application of the Open Platform.

Data reception requires using the [System Data Subscription Interface](#).

Data adjustment: if the pushed data does not meet your requirements, please contact the technical team.

Data Example

```
[
  {
    "deviceType": "system",
    "systemId": "KXGCS1727160960",
    "snCode": "KXGCS1727160960",
```

```
"value": {  
  "onOffGridStatus": "on",  
  "inverterMaxActivePowerW": "2400",  
  "inverterMaxApprentPowerVar": "2511",  
  "systemStatus": "running",  
  "batteryRatedChargePowerW": "2000",  
  "batteryRatedDischargePowerW": "2000",  
  "gridMaxBackfeedPowerW": "2000",  
  "inverterMaxAbsorptionPowerW": "2000",  
  "batteryRatedCapabilityWh": "4000"  
}  
}  
]
```

Data Field Description

Name	Type	Description
systemId	String	Unique code of the power station
snCode	String	Device serial number code
deviceType	String	Type of device
value	String key-value	Telemetry data

For detailed signal description, please refer to [System Data Subscription Signals](#)

Reminders

Relevant data will be pushed when alarm generation.

Data reception requires using the [Alarm Subscription](#)

Data Example

```
[
  {
    "systemId": "KXGCS1727160960",
    "alarmCode": "1001",
    "status": "generation",
    "changeTime": 1716173149647
  }
]
```

Data Field Description

Name	Type	Description
systemId	String	System ID.
alarmCode	Integer	Code of the alarm.
status	String	Status for alarm: generation/recovery.
changeTime	Long	Time of event occurrence.

During the operation of the system, in case of any severe failures that seriously affect system stability, the monitoring and alarm functions we provide will play a crucial role. These functions are designed to offer vendors a convenient system information monitoring service, enabling them to obtain real-time system information at any time through compatible terminals.

If you want to know the detailed content of monitoring signals, such as the specific other types supported, how to customize monitoring signals, or if you have any questions during the use process, please contact our professional support team.

Inverter Alarm Code

Alarm code	Description	Severity Level
1001	Software version mismatch.	Critical
1002	Low insulation resistance.	Critical
1003	The temperature is too high.	Critical
1004	Equipment failure.	Critical
1005	The system grounding is abnormal.	Critical
1006	PV string voltage is high.	Critical
1007	PV string reverse connection.	Critical
1008	PV string back-filling.	Critical
1009	AFCI fault.	Critical
1010	Grid outage.	Critical

Alarm code	Description	Severity Level
1011	Grid overvoltage.	Critical
1012	Grid undervoltage.	Critical
1013	Grid overfrequency.	Critical
1014	Grid underfrequency.	Critical
1015	Grid voltage imbalance 1.	Critical
1016	The DC component of the output current exceeds the limits.	Critical
1017	The leakage electricity exceeds the Limits.	Critical
1018	Communication abnormal.	General
1019	System internal protection.	Critical
1020	AFCI self-test circuit fault.	Critical
1021	Off-grid protection.	Critical

Battery Alarm Code

Alarm code	Description	Severity Level
2001	Software version mismatch.	Critical
2002	The energy storage module has low insulation resistance to ground.	General
2003	The temperature is too high.	Critical
2004	Equipment failure.	Critical

Alarm code	Description	Severity Level
2005	Below desired temperature.	Critical
2008	System internal protection.	Critical

Gateway Alarm Code

Alarm code	Description	Severity Level
3001	Software version mismatch.	Critical
3002	The temperature is too high.	Critical
3003	Equipment failure.	Critical
3004	Excessive leakage current in off-grid output.	Critical
3005	N line grounding fault.	Critical

Currently, our system supports various commands, such as modifying the energy storage operating mode and issuing battery instructions. For battery commands, Sigenergy supports multiple scenarios to meet different application contexts and personalized requirements.

During operation, parameter settings are crucial. In forced charge or discharge modes, the power must be specified, as the system requires a clear power value to drive the equipment; without specifying the power, the system cannot respond to the command. In other modes, setting the power is not required. For specific usage, please refer to the examples.

Query the current energy storage operating mode of a power station, only available in Northbound mode.

Access Restrictions

Each account can access a single power station only once every five minutes.

Request Parameters

Name	Type	Description
systemId	String	Unique identifier of the system

Response Information

Name	Type	Description
code	Integer	Error code
msg	String	Message
data	Object	Detailed setting information as follows
>energyStorageOperationMode	Integer	Energy storage operating mode. Refer to Operational Mode Enum

Successful Response Example

```
{  
  "code": 0,  
  "msg": "success",  
  "data": {
```

```
    "energyStorageOperationMode": 0  
  }  
}
```

Failed Response Example

```
{  
  "code": 1000,  
  "msg": "param illegal",  
  "data": false  
}
```

Set the energy storage operating mode of the corresponding system, only available in Northbound mode.

Notes

Data exchange uses JSON format.

In the Northbound scenario, this interface requires authorization.

Request Parameters

Name	Type	Required	Description
systemId	String	Yes	Unique identifier of the system
energyStorageOperationMode	Integer	No	Energy storage operating mode. Refer to Operational Mode Enum

Request Example

```
{
  "systemId": "NDXZZ1731665796",
  "energyStorageOperationMode": 0
}
```

Switch Operating Mode — MQTT version, supports switching the operating mode of the energy storage system.

Notes

Data exchange uses JSON format.

In the Northbound scenario, this interface requires authorization.

Request Format

Name	Type	Required	Description
accessToken	String	Yes	Authorization token obtained from the Chapter 2 interface
systemId	String	Yes	Unique identifier of the power station
mode	String	Yes	Energy storage operating mode. Refer to Operational Mode Enum

Request Example

```
{
  "accessToken": "JFf_QTaUkGTD9AQXiMrfLBUfM3v2qyLPr3K0o1e",
  "systemId": "CBFGA1627168228",
  "mode": "NBI"
}
```

This API is used to send control commands to a specified energy storage system, enabling efficient management and scheduling of the system's energy. Through this interface, users can flexibly control the charging and discharging states, power limits, and PV/grid priority strategies of the energy storage system to meet different operational scenarios. Typical applications include:

Battery Operation Mode Control: Set the system to charge, discharge, self-consumption, or idle modes to adapt to various energy management strategies.

Power Limit Configuration: Set charging power, PV charging power, maximum export power, and maximum import power to finely control the energy inflow and outflow of the system.

Priority Strategy Management: Configure the source priority for charging and discharging (PV or grid/battery), supporting flexible scheduling and optimized energy utilization.

Time-Slot Control: Specify the start time and duration of commands to achieve scheduled operations and batch dispatch.

Batch Command Issuance: For each site, a maximum of 24 instructions can be received in each batch.

Request Style

Parameter	Type	Required	Description
accessToken	String	Yes	Authorization token obtained from Chapter 2
systemId	String	Yes	Unique code of the power station
activeMode	String	Yes	System active mode
startTime	Long	Yes	Command start time, in seconds

Parameter	Type	Required	Description
duration	Integer	Yes	Command duration, in minutes
chargingPower	Double	No	Max energy storage charging/discharging power (KW)
pvPower	Double	No	Max photovoltaic charging power (KW)
maxSellPower	Double	No	Max export power to the grid (KW)
maxPurchasePower	Double	No	Max purchase power from the grid (KW)
chargePriorityType	Enum	No	Charging priority (PV/GRID)
dischargePriorityType	Enum	No	Discharging priority (PV/BATTERY)

Sample Request

Basic Example

```
{
  "accessToken": " JFf_QTaUkGTD9AQXiMrfLBUfM3v2qyLPr3K0ole ",
  "commands": [{
    "systemId": "KXGCS1727160960",
    "activeMode": "charge",
    "startTime": 1715154185,
    "duration": 2,
    "chargingPower": 3.2,
    "pvPower": 1.8
  }]
}
```

Scenario 1: No PV power curtailment, battery charged by PV

```
{  
  "systemId": "FAZGW8745476782",  
  "activeMode": "selfConsumption",  
  "startTime": 1691572800000,  
  "duration": 30  
}
```

Scenario 2: PV power curtailed, battery charged by grid first

```
{  
  "systemId": "FAZGW8745476782",  
  "activeMode": "charge",  
  "startTime": 1691572800000,  
  "duration": 30,  
  "chargingPower": 25.0,  
  "chargePriorityType": "GRID"  
}
```

Scenario 3: PV curtailed, battery idle, grid does not supply power

```
{  
  "systemId": "FAZGW8745476782",  
  "activeMode": "idle",  
  "startTime": 1691572800000,  
  "duration": 30,  
  "maxSellPower": 0  
}
```

Value	Integer Value	Description	Switch to northbound
MSC	0	Maximum Self-Consumption	Yes
FFG	5	Fully Feed-in to Grid	Yes
VPP	6	VPP	/
NBI	8	North Bound	/

Operating Mode Description

Enumerations	Description
charge	Forced battery charging.
discharge	Forced battery discharge.
idle	Not charging and not discharging.
selfConsumption	Surplus PV power gives priority to charging.
selfConsumption-grid	Surplus PV power prioritizes grid injection.

Please refer to [Detailed of Energy Storage Operating Modes](#) for specific logic for each mode.

Charge Priority Description

Enumerations	Description
PV	Prioritize charging from solar PV
GRID	Prioritize charging from grid

Discharge Priority Description

Enumerations	Description
PV	Prioritize discharging PV power
BATTERY	Prioritize discharging battery power

The Openapi system offers ample flexibility to accommodate various signals, instruction modes, and alarm types. Here are some examples supported by Openapi. Additionally, there are detailed introductions to energy storage operation modes below.

Signals And Mode > Telemetry Signals Description

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Name	Description
gridActivePowerW	The active power at the grid point.
gridReactivePowerVar	The reactive power at the grid point.
storageSOC%	The state of charge of an energy storage system, expressed as a percentage of its total capacity.
inverterPhaseAActivePowerW	The active power in phase A.
inverterPhaseBActivePowerW	The active power in phase B.
inverterPhaseCActivePowerW	The active power in phase C.
inverterPhaseAReactivePowerVar	The reactive power in phase A.
inverterPhaseBReactivePowerVar	The reactive power in phase B.
inverterPhaseCReactivePowerVar	The reactive power in phase C.
inverterActivePowerW	The total active power output of all inverter.
inverterReactivePowerVar	The total reactive power output of all inverter.
pvPowerW	The total power output of the photovoltaic (PV) system.
storageChargeDischargePowerW	The total power capacity of the energy storage system, where charging is represented as positive and discharging as negative.

Name	Description
gridPhaseAActivePowerW	The active power in phase A at the grid connection point.
gridPhaseBActivePowerW	The active power in phase B at the grid connection point.
gridPhaseCActivePowerW	The active power in phase C at the grid connection point.
gridPhaseAReactivePowerVar	The reactive power in phase A at the grid connection point.
gridPhaseBReactivePowerVar	The reactive power in phase B at the grid connection point.
gridPhaseCReactivePowerVar	The reactive power in phase C at the grid connection point.
storageChargeCapacityWh	The amount of energy that an energy storage system can be charged with.
storageDischargeCapacityWh	The amount of energy that an energy storage system can discharge.
batteryMaxChargePowerW	The maximum power all system batteries can collectively receive during charging.
batteryMaxDischargePowerW	The maximum power all system batteries can collectively deliver during discharging.
inverterMaxFeedInActivePowerW	The combined maximum active power capacity of all inverters within the system.
inverterMaxAbsorptionActivePowerW	The total active power absorption capacity of all inverters within the system.

Name	Description
inverterMaxFeedInReactivePowerVar	The maximum reactive power that all inverters in the system can collectively feed into the system.
inverterMaxAbsorptionReactivePowerVar	The total reactive power absorption capacity of all inverters within the system.

Signals And Mode > System Signals Description

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Name	Description
onOffGridStatus	The system is on grid or off grid.
inverterMaxActivePowerW	This is should be the base value of all active power adjustment actions.
inverterMaxApprentPowerVar	This is should be the base value of all reactive power adjustment actions.
systemStatus	This should be the status of the system. Enumeration: standby running fault shutdown disconnected
batteryRatedChargePowerW	Rated energy storage charging power is the maximum power a battery can safely receive during charging.
batteryRatedDischargePowerW	Rated energy storage discharging power is the maximum power a battery can safely deliver during discharging.
gridMaxBackfeedPowerW	Permissible feed in power at the grid connection point is the maximum power that is allowed to be sent from the grid into a local system.
batteryRatedCapabilityWh	The total rated capacity of system refers to the maximum amount of electrical energy the entire system can generate or store under specific operating conditions.
inverterMaxAbsorptionActivePowerW	The highest active power that can be absorbed from the grid by a system.

Name	Description
chargeCutOffSOC%	Charge cut-off SOC. When battery SOC $\geq$ this threshold, the system stops charging. Range: 0–100.
dischargeCutOffSOC%	Discharge cut-off SOC. When battery SOC $\leq$ this threshold, the system stops discharging. Range: 0–100.
backupCutOffSOC%	Backup cut-off SOC. In backup mode, when battery SOC $\leq$ this threshold, discharging is limited or stopped. Range: 0–100.
peakShavingCutOffSOC%	Peak shaving cut-off SOC. Effective only when peakShavingStatus $\neq$ 0; when battery SOC $\leq$ this threshold, peak shaving discharge is stopped. Range: 0–100.
peakShavingStatus	Peak shaving status. Indicates whether the system is in peak shaving control mode and affects charge/discharge behavior. Enumeration: off = Disabled; on = enabled
stormWatchStatus	Storm Watch status. Indicates whether the system is in Storm Watch mode and affects charge/discharge behavior. Enumeration: off = Disabled; on = enabled

Charge

In this mode, the system will prioritize using photovoltaic (PV) energy to charge the energy storage system, followed by grid energy. When there is sufficient solar energy, the system will use PV energy exclusively to charge the energy storage system. Any excess PV energy will be fed into the grid. If solar energy is insufficient, the system will use solar energy for charging while simultaneously drawing energy from the grid.

When using this mode, if it is necessary to limit the power purchased from or fed back to the grid, sensors must be connected to the grid-feeding points:

If the grid purchasing power limit is set, during charging, the system will not exceed the specified grid purchasing power;

If the grid-feeding point power limit is set, the system will not exceed this reverse power when feeding solar power to the grid.

Discharge

In this mode, the system will prioritize using photovoltaic (PV) energy for discharge, followed by energy from the storage system.

When using this mode, if it is necessary to limit the backflow power at the grid connection point, a grid connection point sensor must be connected. If a backflow power limit is set, the system will not exceed this backflow power when feeding PV power into the grid.

Self-Consumption

In this mode:

When solar energy is abundant, the electricity generated by the photovoltaic system will be prioritized for supplying the load. Any surplus electricity will be stored in the energy storage system, and the excess will be sold to the grid;

When solar energy is insufficient, the energy storage system will discharge electricity to supply the load.

Self-Consumption Grid

In this mode:

When there is an abundance of solar energy, the electricity generated by the photovoltaic system will be preferentially supplied to the load, and any surplus electricity will be sold to the grid;

When solar energy is insufficient, the energy storage system will discharge electricity to supply the load.

Idle

In this mode, the battery neither charges nor discharges. The surplus electricity from the PV system is exported to the grid.

When using this mode, if it is necessary to limit the power fed back to the grid, sensors must be connected to the grid-feeding points. If the grid-feeding point power limit is set, the system will not exceed its maximum allowable power when feeding solar power to the grid.

Error Code List

This chapter lists the error codes that may be returned during the interface call of Sigenergy Openapi and their corresponding description information, which is convenient for service providers to quickly locate problems, troubleshoot faults, and ensure the smooth interface interaction process.

NO.	Error Code	Description
1	0	Success
2	1000	Param illegal
3	1101	Wrong serial
4	1102	Registration incomplete
5	1103	In other VPP
6	1104	Device offline
7	1105	Current software version does not support VPP
8	1106	Station was not found
9	1107	AIO units and Inverters only
10	1108	Station info not found
11	1109	RPC fail
12	1110	Interface is current-limited
13	1111	Station not permitted

NO.	Error Code	Description
14	1112	In other VPP (Evergen)
15	1201	Access restriction
16	1301	Client not found
17	1302	Station status anomaly
18	1303	Client has existed
19	1304	Firmware version mismatch
20	1401	No permission to operate this station
21	1402	No permission
22	1501	Failed to execute command
23	1502	Sigenergy system internal error
24	1503	The power station has the anti-backflow setting enabled
25	1504	The power station has enabled peak shaving
26	1600	The invitation is invalid
27	1601	Account system error
28	1602	Account already registered
29	1603	Account unReviewed
30	1604	Developer not approved